The Chinese University of Hong Kong
Department of Computer Science and Engineering
CSCI3170 Introduction to Database Systems

2026-27 Term One Assignment 1:
Conceptual and Logical Data Modelling + Relational Algebra + Basic SQL

Release Date: October 7, 2026 (Wednesday Week5)
Due Date: **October 14, 2026, at 11:59pm**. (Wednesday Week 6)
Total marks available: 93 marks

Please submit your assignment onto the Blackboard system. No submission will be accepted past October 16, 2026 at 11:59pm. Refer to Week One Notes for more penalty related information.

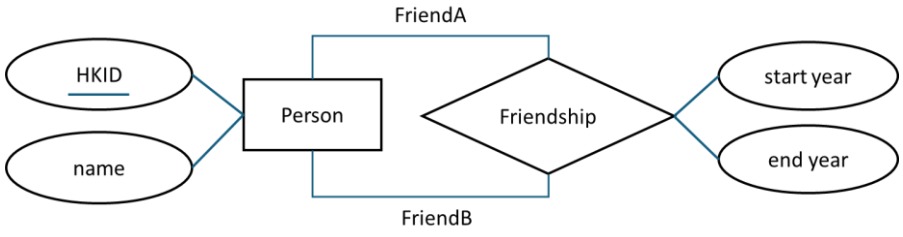
- Where I am aware that AI tools exist, please do these questions yourself, they each target exploring some interesting concepts, and make for a valuable learning experience.
- Use the course notation (use arrows, rather than the 1, m on the edges).
- Each question is worth 3 marks. If a question does not have components, the question as a whole is trivially worth 3 marks. (There are 26 questions total, sub-questions in question 6 is worth 6 marks each)

Additional Instructions,

- *Start and submit each section on a separate **.pdf** document file.*
  a. *<cuhkID>_ass1_sectA.pdf*
  b. *<cuhkID>_ass1_sectB.pdf*
  c. *<cuhkID>_ass1_sectC.pdf*
- *Above instructions not followed will result in minor deductions.*
- *Handwritten solutions are recommended to prepare for quiz/exam conditions; however not mandatory.*

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Section A – E/R Modelling + Translation to Relational Model
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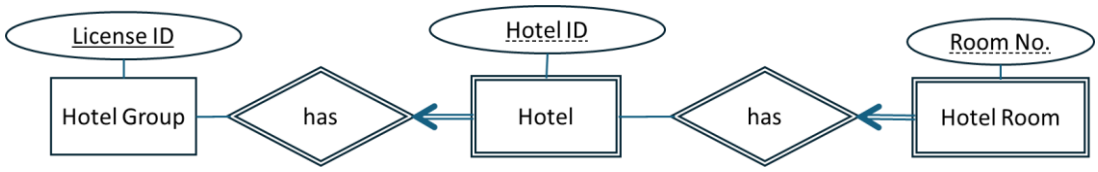
1. Translate the E/R diagram of a recursive relationship below in attempt to model a symmetric concept.



a. Draw the final translated relational diagram.
b. Does model allow us to query information such as “friends of a friend who is also my friend”?
c. Based on the the resulting logical model… Identify and briefly explain two real limitations on modelling the concept of friendship as such.

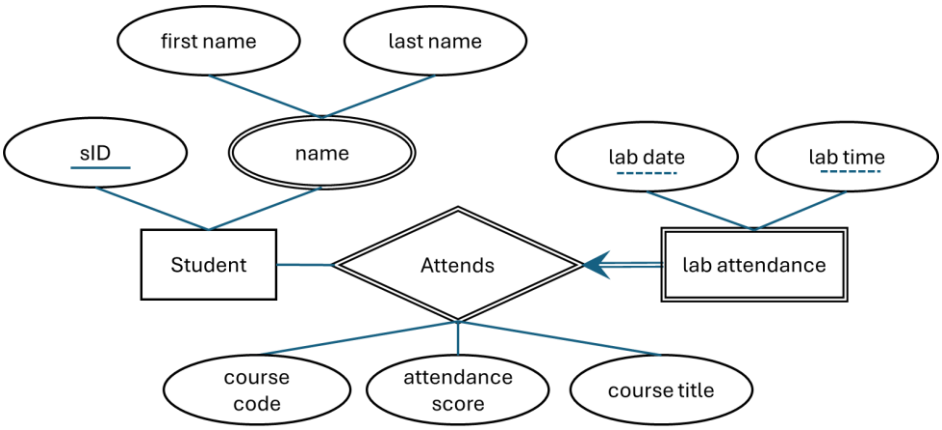
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2. In relation to the following Entity-Relation Diagram, answer relevant questions.



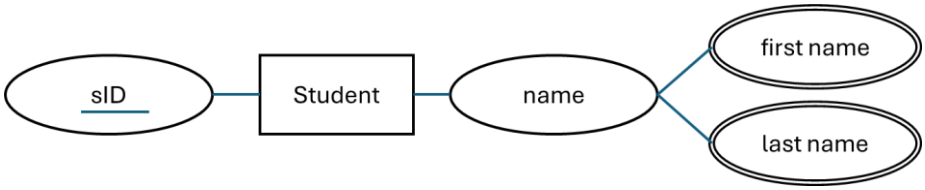
a. Translate above E/R diagram into a Relational diagram.
b. A student argues that, as an ID, any Hotel ID must only ever be unique and thus insists on modelling it as a strong entity instead. Comment on whether it is still the same scenario if Hotel were to be modelled instead with a strong entity.

3. Consider the E/R diagram provided here ***[Note: inspired from my 24-25(T2) CSCI3170 Exam]***



a. If the goal was to also retain course entity related information, based on your knowledge on translations, briefly whether you believe this E/R model here effectively achieves this goal.
b. Translate the E/R diagram above, draw the final translated relational diagram.

Suppose that an intern proposed the following alternative conceptual model for the entity set Person. Note: to avoid confusion, let the entity set *Student* from [q3(b)] be referred to as '*StudentA'*, and the entity set *Student* here be referred to as '*StudentB'*.



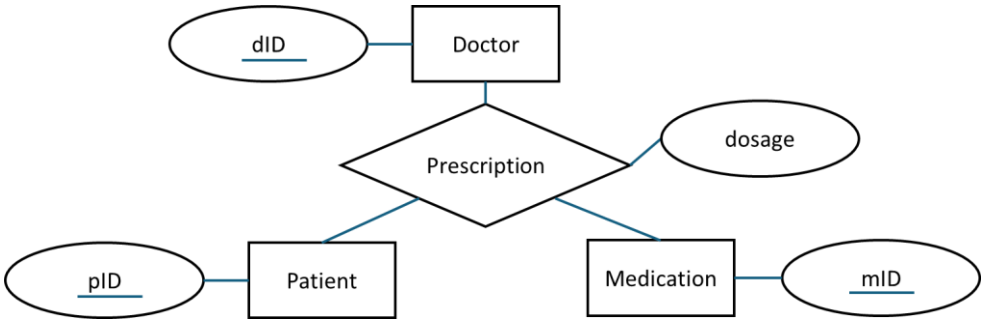
c. Translate the E/R diagram for the entity set '*StudentB'* above, draw the final translated relational diagram.
d. Clearly discuss the semantic difference between the two conceptual models and explain why one of them can be problematic with a concrete example.

Suppose that client is confused by weak entities and prefers not to use them. Your supervisor proposes that you remodel "lab attendance" as a multi-valued attribute for entity-set *Student*.

e. Redraw the new E/R-diagram with this change whilst retaining all existing information
f. Redraw a new relational diagram with the above updated E/R-diagram.
g. Comment on one logical difference in implications between lab attendance in your relational diagrams from [q3(b)] and that from [q3(f)] and explain whether or not one implementation may be more preferable than the other.

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4. You are given the following entity relationship schema, where the primary keys are underlined. A ternary relation has been selected to express the complex relationship between doctor(s), patient(s), and medication(s).



After you translate it into a logical schema, the clinic adds some initial data as shown below.

Prescription (<u>D</u>, <u>P</u>, <u>M</u>, dosage):
(D1, P1, M1, 10mg)
(D1, P2, M2, 5mg)
(D2, P1, M2, 20mg)

For this statement (1) answer Truth or False, and (2) provide a very short and simple explanation/example. ***Note****: A correct Truth or False statement is given one point, on that basis, two additional points are given for your explanation/example.*

a. If we include dosage as a key-attribute, we can still store the same data/ information as when we do not include it as a key-attribute.

Suppose that you remember from CSCI3170, that binary relations are strongly recommended. Your staff had "decomposed" this ternary relation into three binary relations – one relation between each possible pair of entity-set). One of the relations is **Treats (D, P)**, has been updated to preserver the following information from the *Prescription* relation.

(D1, P1)
(D1, P2)
(D2, P1)

b. Similarly, what would be the corresponding records for the remaining two binary relations?
Authorizes (D, M)
Takes (P, M)

c. Suppose all three binary relations are now created, and the original ternary relation *"Prescription"* is permanently deleted. Explain clearly with one or two concrete example(s) any issue(s) caused by this decomposition. *Hint: It is sometimes said that ternary relations capture information that pairwise relations cannot preserve without reintroducing the full triple.*

d. Provide an alternative ER-diagram that (1) solves the issue you listed above, (2) does not use any n-ary relationships where n is 3 or above, and (3) effectively preserves the initially intended initial prescription information.

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Section B – Relational Algebra and simple SQL
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5. You are given the following employee database schema, where the primary keys are underlined. Please formulate the appropriate relational algebra expressions. You **may** use ≠ in the any of your relational algebra predicates.

Customer(cid, name, address)
Product(pid, ptype, price)
Purchase(cid, pid, date)

Details:

- *Cid of Buy is a foreign key referring to cid of Customer.*
- *Pid of Buy is a foreign key referring to pid of Product.*

a. Find the *cid(s)* of customer who have never bought chairs.
b. Find the *cid(s)* and name of customer have never bought anything.
c. Find the *name(s)* of customer who have bought exactly one chair.

6. You are given the following database schema, where the primary keys are underlined. Please formulate the appropriate expressions. You **may not** use ≠ in the any of your relational algebra predicates. Answer the questions once in Relational Algebra, and once more using SQL

Flights(flno, from, to, distance, departs, arrives)
Aircraft(aid, amanufacturer, cruisingrange)
Employees(eid, ename, salary)
Certified(eid, aid)

Details:

- *Employees relation describes pilots and other kinds of employees as well.*
- *Every pilot is certified for some aircraft (otherwise, they are not pilots)*
- *Only pilots are certified to fly*
- *aid of Certified is a foreign key referring to aid of Aircraft.*
- *eid of Certified is a foreign key referring to eid of Employees.*
- *An Aircraft can be used for non-stop flights if its crusingrange is greater than the flight distance*

a. Find the *flno(s)* of flights that fly to HK.
b. Find the *aid(s)* of the aircraft(s) with the largest cruising range.
c. Find the *eid(s)* of pilots certified for aircrafts with cursing range not less than 8,000(unit: miles).
d. Find the *name(s)* of pilots who can fly direct flights from New York to Los Angeles but are not certified on any Airbus aircraft.
e. Find the *salaries* of pilot(s) who are certified to fly exactly three Boeing aircrafts.

7. You are given the following employee database schema:

Employee(id, name, street, city)
Works(id, company-name, salary)
company(company-name, city)

a. What does the following relational algebra expression do?

$$\Pi_{ID,person_name}(employee) - \Pi_{A.ID,A.person_name}(\rho_A(employee) \bowtie_{A.salary<B.salary} \rho_B(employee))$$

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Section C – There are no marks to earn from the following questions.

- **Attempt or not, these have zero advantage/ penalization towards your assignment grade**
- New class of questions present to highlight scenarios and test application and proficiency.
- Contents taught thus far will be minimally sufficient to resolve these (ungraded) questions.

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8. Referring to Question 6. For a plane with aid = 809, find all the destinations it can fly to from Hong Kong that is only available via two-hop (I.e., Hong Kong to A, then from A to B), and is still within its maximum cruising range. We assume stopping at an airport includes no refuelling, and doesn't reset remaining cruising range. Solve once with SQL, and (try) once with Relational Algebra.

9. Explore the following scenario seriously and propose your best resolution to a satisfactory degree.

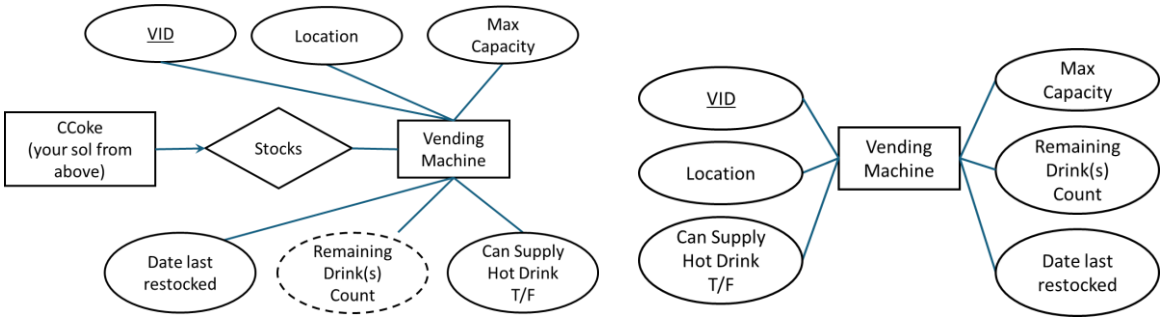
Even with surrogate key… considering the case of the CCoke cans shown as early as Lecture One, two instances of the same entity can effectively be practically the "same" (or very hard to distinguish). [SEE APPENDIX BELOW]. Given this, you are recruited by the CCoke company that manages multiple factories and multiple stores, etc. Your mission is to conceptually model this scenario such that the following data requirement is satisfied *"each code can be uniquely referred-to / identified";* where you are able to uniquely (and still reasonably) identify two individual cans of CCoke effectively within your database system up until the moment it falls in the hands of the consumer. (Hint: see question 2)

Task: You may explore options or related entities, draw one final E/R-diagram, and its equivalent relational diagram. state your assumptions if any, and any known limitations that you know about your conceptual modelling.

I. Please keep to using only the mechanisms defined in our basic E/R model, discussed in class
II. You are only allowed to refer to attributes or relationships that are present in the real world (zero surrogate keys)
III. For this question only, if your work for this question was a collaboration with other students, please include their name and Sid here for my reference

10. Suppose you are then also invited to expand your database scenario to include the many vending machines (connected to the internet) you manage. To simply the scenario, each machine has only one button that always sells only one drink - CCoke.
Rather than restocking periodically; they wish to restock them if they fall below a number. Thus, they are keen to store, and thus keep track of remaining drinks in each machine.
Your friend proposes two models below:

a. Taking into account the two resulting logical model(s), identify the key difference(s).
b. Comment on whether this issue is different in natural to the textbook case of *age*.
c. Decide on whether you would treat *remaining-drink-count* as derived attribute or a non-derived attribute. Justify your decision.
d. Given the above assumptions, why not add one attribute *name-of-drink-sold* to the entity?



End of 2627t1 Ass One, Well done.

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APPENDIX:



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